

- 3 (a) loss in potential energy due to decrease in height (as P.E. = mgh) (B1)
gain in kinetic energy due to increase in speed (as K.E. = $\frac{1}{2}mv^2$) (B1)
special case 'as PE decreases KE increases' (1/2)
increase in thermal energy due to work done against air resistance (B1)
loss in P.E. equals gain in K.E. and thermal energy (B1)
max. 3 [3]

Page 3	Mark Scheme	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2013	9702	21

- (b) (i) kinetic energy = $\frac{1}{2}mv^2$ C1
 = $\frac{1}{2} \times 0.150 \times (25)^2$ C1
 = 46.875 = 47 J A1 [3]
- (ii) 1. potential energy (= mgh) = $0.150 \times 9.81 \times 21$ C1
 loss = KE – mgh = $46.875 - (30.9)$ C1
 = 15.97 = 16 J A1 [3]
2. work done = 16 J
 work done = force $\times$ distance C1
 $F = 16 / 21 = 0.76$ N A1 [2]